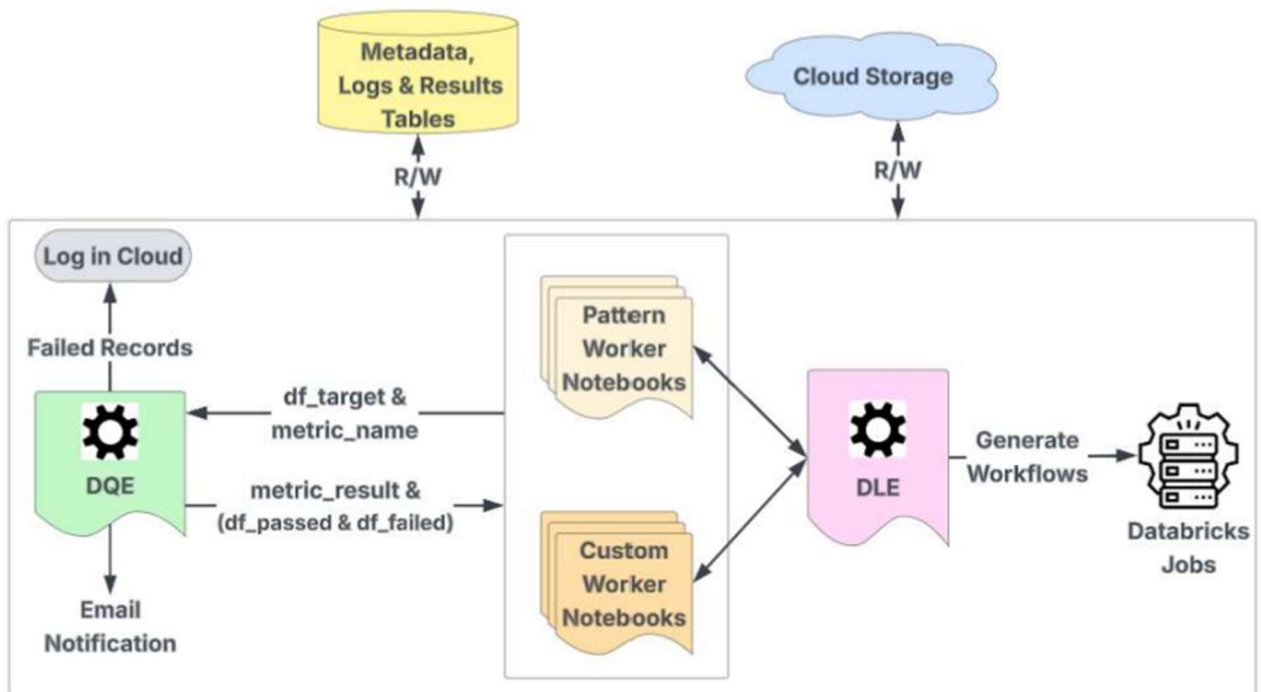


Designed and Developed two Major Products for ETLs: DLE (Data Load Engine) & DOE (Data Quality Engine)

Designed and implemented two metadata-driven frameworks to manage ETL processes across all data layers, supporting both data loading operations and comprehensive data quality validation checks.



- **DOE Rules:** NOT_NULL, UNIQUE, REF_INT, SQL_FILTER, CUSTOM_FUNC & etc..
- **Action on Failure by DQE:** ALERT | QUARANTINE | BLOCK
- **DLE Loader:** Pattern-Based / Customer Logics